

Adaptive Experimental Design for Detecting Hidden Robustness in AI Alignment Interventions

Phase: experiment_runner (Phase 2) · Date: 2026-08-05 · Compute: CPU for E1?E3; 2× NVIDIA RTX A6000 for E4 (~10 GPU-min per model)

1. Executive Summary

Research question. When an alignment evaluation returns a null ? "we could not elicit the
Key finding, in one sentence. Adaptivity helps a great deal, but almost none of the
The result decomposes into six findings, four of which are compute-matched comparisons:
1. The motivating phenomenon is real and extreme. The default design (plain_init
Practical implication. The actionable advice for a safety team is not "instrument your

2. Research Question & Motivation

The hypothesis under test, verbatim from the brief:
> Adaptive experimental protocols that use null results as iterative triggers for
The motivating ambiguity is that a null can mean either genuine robustness or an
The gap. Adaptivity has been argued for attacks (Andriushchenko et al. 2024) and for

3. Methodology

3.1 Substrates

| Experiment | Substrate | Size | Role |

Nothing in E1?E3 is re-run or simulated: these are the original authors' own traces and the

3.2 The core methodological device: matched-budget replay

Every experiment follows the same pattern. An outcome matrix is evaluated (or read) once,

3.3 Protocol arms

E1 (within-search escalation ladder, Llama-2-7B-chat, 3 design families):

| Arm | Definition |

LADDER is constructed so that disabling the signal (? = 0, extend = 1) recovers
E2 (portfolio allocation): STATIC-single-method (the field default), STATIC-round-robin
E3 (calibration of the terminal null): FIXED-t CP (Clopper-Pearson at a pre-registered

3.4 Statistical analysis (pre-registered in planning.md before running)

- Bootstrap over behaviours, 1,000 resamples, seed 42. Never from re-running attacks ?

3.5 Reproducibility

Seed 42 everywhere; E4 decoding is greedy (do_sample=False, temperature 0). Environment:

4. Results

4.1 E1 ? the escalation ladder at matched iteration budget

![E1](figures/e1_detection_vs_budget.png)

Detection rate on 50 Llama-2-7B-chat behaviours (n = 50, 95% bootstrap CI):

| Budget | STATIC-d0 | STATIC-d1 | STATIC-d2 | FIXED-SPLIT | LADDER | ORACLE |

Two things happen at once.

The static-design failure is total. STATIC-d0 detects nothing at any budget across four
But the null-trigger adds nothing. Against FIXED-SPLIT ? same budget, same three
Against STATIC-d0, by contrast, LADDER wins overwhelmingly (B = 60: ? = +0.86, 95% CI

A $3 \times 3 \times 3$ sweep over the trigger's hyper-parameters (patience $\in \{2,5,20\}$, $\gamma \in \{0.1, 0.5, 1\}$)

4.2 E1c γ why the trigger fails: nulls are not diagnostic

!{E1c}(figures/e1c_null_diagnosticity.png)

Conditional on a run still being a null at iteration t γ precisely the decision context an

| t | n | configs | pooled AUC | 95% CI | base rate |

Every confidence interval covers 0.5. The surrogate signal that the whole "null as

4.3 E2 γ portfolio-level adaptivity does work

!{E2}(figures/e2_portfolio_allocation.png)

Fraction of 100 behaviours jailbroken at matched budget (300 policy replicates per cell):

GPT-4-0125-preview (design-space ceiling 0.86):

| Budget | STATIC-single | STATIC-round-robin | STATIC-best(oracle) | ADAPT-thompson | ADAPT-escal+thomp | ORACLE |

GPT-3.5-turbo-1106 (ceiling 0.95):

| Budget | STATIC-single | STATIC-round-robin | STATIC-best(oracle) | ADAPT-thompson | ADAPT-escal+thomp | ORACLE |

Adaptive allocation beats the diversified static schedule by +0.648 on GPT-4 and

The result is not universal, and the exception is informative. On Llama-2-7B-chat and

4.4 E3 γ the nulls an adaptive protocol emits are badly calibrated

!{E3}(figures/e3_null_calibration.png)

Empirical coverage of a nominally 95% upper bound on the elicitable fraction, over 14

| Method | fraction defined | coverage | mean bound | mean truth |

Restricted to evaluators that stopped on a total null (the case that matters most for

Two conclusions follow, and the second is the one worth carrying:

1. All four bounds under-cover badly. A nominally 95% interval delivers $\sim 64\%$.

4.5 E4 γ live replication on real open-weights LLMs

Three real open-weights instruct models were run locally (greedy decoding, seed 42) on 32

!{E4 families}(figures/e4_family_flip_rates.png)

Flip rate by design family (intervention active; n = eligible items):

| Model | n | P1 simple doubt | P2 authority | P3 fabricated consensus | P4 persistent multi-turn | ceiling |

This is the hypothesis's motivating scenario reproduced live and from scratch. **A static

The static protocol also mis-measures the intervention itself. Judged by P1 alone, the

!{E4 detection}(figures/e4_null_to_nonnull.png)

Null-to-non-null transition rate γ P(the protocol reports at least one flip), the

| Budget | Qwen2.5-7B
STATIC-P1 / round-robin / ADAPT-thompson | Phi-3.5-mini
STATIC-P1 / round-robin / ADAPT-thompson |

STATIC-P1 on Phi-3.5-mini never escapes the null at any budget up to 200 calls, while

E4 independently reproduces E2's pattern on live models, not replayed logs: adaptivity

5. Analysis & Discussion

5.1 Verdict on the hypothesis

The hypothesis has three claims. They receive different verdicts.

| Claim | Verdict | Evidence |

The honest summary is that the hypothesis is **right about the conclusion and wrong about

5.2 Why the two levels differ

The two levels of adaptivity face different signal-to-noise regimes. At the portfolio

This also reconciles our result with Andriushchenko et al., whose headline (0% γ 50% γ 100%

5.3 Error analysis and failure modes

- When adaptivity loses (E2, Llama-2/Vicuna): a dominant method exists, so exploration is

5.4 Surprises

The genuinely unexpected result is E3's ordering: we expected anytime-valid inference to

6. Limitations

1. Retrospective replay. E1?E3 replay logs generated by other people's design choices.

7. Conclusions & Next Steps

Answer to the research question. Adaptive protocols do find substantially more at matched

Recommendations for practice.

Next experiments. (a) Test whether a better within-run signal (true elicitation

8. References

- Andriushchenko, Croce, Flammarion (2024). Jailbreaking Leading Safety-Aligned LLMs with Simple Adaptive Attacks. arXiv:2404.02151 ? source of rs_trajectories.

Artifacts. results/offline/ (E1?E3 raw + summaries + tests), results/live/ (E4 grid